Seminar 2 - Lectures 4 and 5
Computer Vision 1, Master AI, 2026

Slides and Exercises inherited from Arnoud Visser, Martin Oswald, Qi Bi, Roan van Blanken

1 Exercise 1: Mathematical Morphology

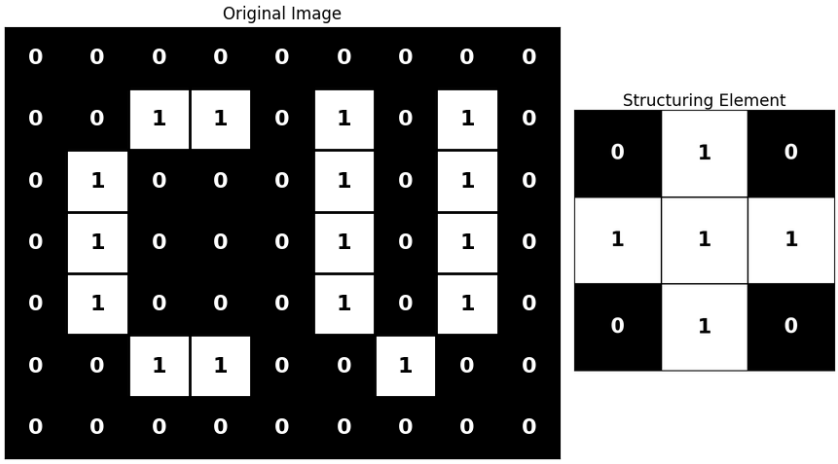


Figure 1: Input binary image and structuring element.

Q.1.a Consider a 7x9 binary image shown in Figure 1, where the value '1' represents the foreground, and '0' represents the background. The 3x3 matrix next to the image is the structuring element.

Perform a dilation operation on the image using the given structuring element, and assume zero padding around the edges of the image.

Q.1.b Consider the same 7x9 binary image shown in Figure 1. Perform an erosion operation on the image using the given structuring element, and assume zero padding around the edges of the image.

Q.1.c Consider the same 7x9 binary image shown in Figure 1. Perform a closing operation on the image using the given structuring element, and assume zero padding around the edges of the image.

Q.1.d Would the result of the closing operation in the previous question have been different if a different padding method (such as mirror, clamp, or wrap) had been used instead of zero padding? Explain your reasoning.

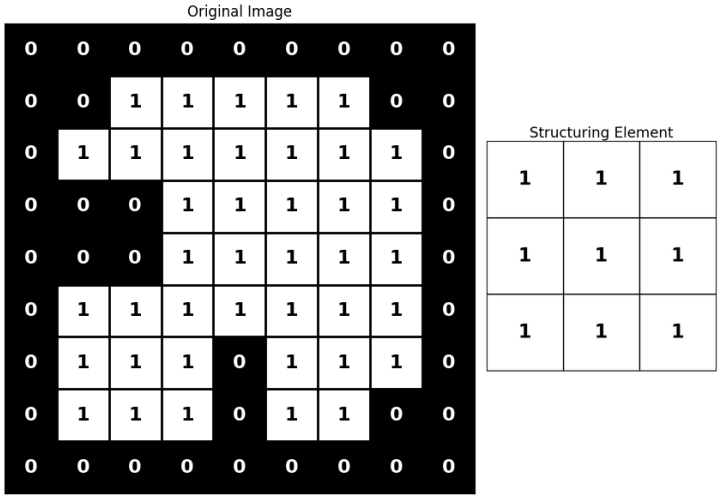


Figure 2: Input binary image and structuring element.

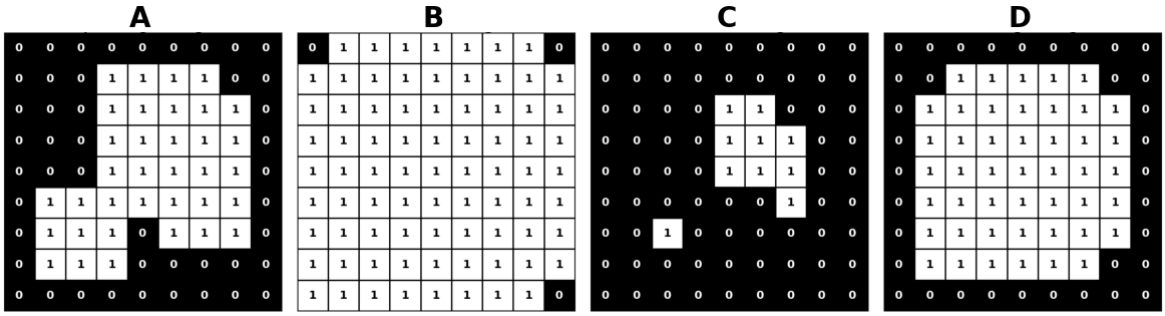


Figure 3: Morphological operation results on the input binary image.

Q.1.e Consider a different 9x9 binary image and structuring element, as shown in Figure 2. Four different morphological operations (Dilation, Erosion, Opening, and Closing) have been applied to this image using the given structuring element. Assume zero padding around the edges of the image.

The results of these operations are shown in Figure 3, with the outputs labeled A, B, C, and D. Match each output to the correct morphological operation.

2 Exercise 2: Connected Component Analysis

Connected Component Analysis, or more broadly, Connected Component Labeling, is a set of algorithms used to identify and label (sub)sets of connected components within a graph. This concept can be naturally applied to an image by treating the image as a graph.

Q.2.a How would you construct an image graph? How are pixels connected?

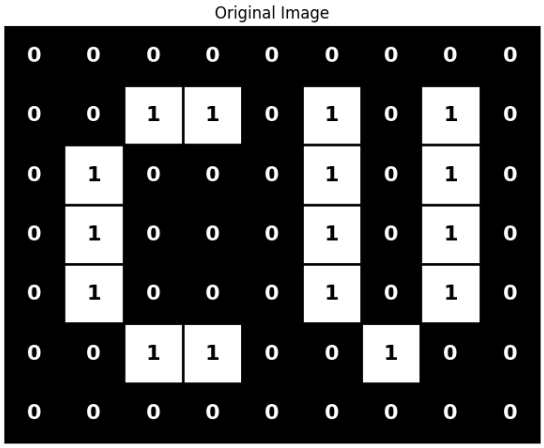


Figure 4: A binary image for connected component analysis.

Q.2.b Identify and label the connected components in the binary image shown in Figure 4. First, use 4-connected neighbors, then use 8-connected neighbors.

3 Exercise 3: Image Filtering

In image processing, kernels are often applied to an image to alter its appearance—a process known as image filtering. A kernel is a local operator that depends only on the neighborhood of pixels around each point. Below are some types of commonly used kernels.

3x3 Uniform (Box) Kernel

$$T = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \underline{1} & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

5x5 Uniform (Box) Kernel

$$U = \frac{1}{25}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & \underline{1} & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Another 3x3 kernel

$$V = \frac{1}{6}\begin{bmatrix} 1 & 0 & -1 \\ 1 & \underline{0} & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Yet another 3x3 kernel

$$W = \begin{bmatrix} 1 & -2 & 1 \\ -2 & \underline{4} & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

3x3 Laplacian Kernel

$$L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & \underline{-4} & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

3x3 Sobel Kernel

$$S = \begin{bmatrix} 1 & 0 & -1 \\ 2 & \underline{0} & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

Q.3.a Which of the following kernels would make an image blurrier: a 3x3 uniform kernel or a 5x5 uniform kernel? Explain your reasoning.

Q.3.b What types of image features are detected by the 3x3 kernels V and W? Describe their functionality.

Q.3.c How can the transformation of an image, achieved by convolving a 3x3 averaging **kernel** T followed by the 3x3 **kernel** W, be implemented using a single 5x5 kernel? Calculate the elements of this resulting kernel.

Q.3.d Demonstrate that the Sobel filter S can be separated into two different 1-dimensional filters. Identify each of these 1-dimensional filters as either a low-pass filter, a high-pass filter, or a band-pass filter.

Consider the following image matrix A:

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Q.3.e Correlate Image A with kernel T. Use zero padding if necessary.

Q.3.f Apply a 3x3 median kernel on Image A. Use mirror padding if necessary.

4 Exercise 4: Hough Transform

The Hough Transform is an image processing technique used to detect instances of shapes within an image. Originally, it was designed to detect lines, but it has since been extended to detect more complex shapes such as circles, ellipses, and arbitrary shapes. For the purposes of these questions, we will focus on the Hough Transform for line detection.

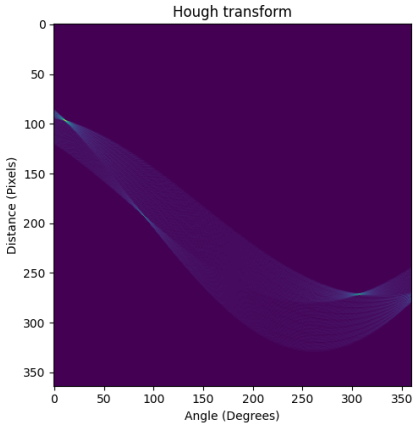


Figure 5: The accumulator matrix depicted as an image. This matrix results from applying a Hough Transform to an image with lines.

Q.4.a Consider the accumulator matrix after applying the Hough Transform to an image, as shown in Figure 5. How many lines are present in the original image?

Q.4.b The accumulator space of the Hough Transform is often expressed in polar coordinates (ρ, θ) instead of line parameters (a, b). Why is this?

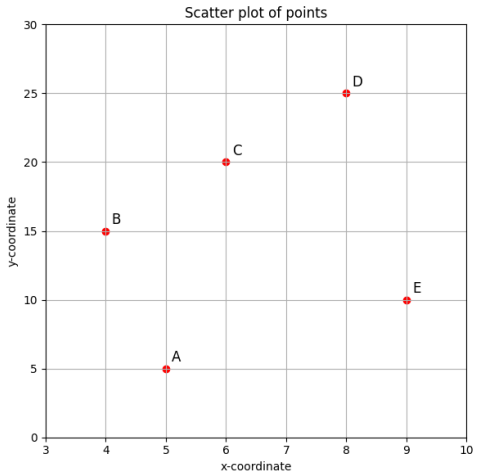


Figure 6: Scatter plot of points

Q.4.c Five points in an image, as shown in Figure 6, are given with the following coordinates: $A = (A_x, A_y) = (5, 5)$, $B = (4, 15)$, $C = (6, 20)$, $D = (8, 25)$, $E = (9, 10)$. Demonstrate how you can fit a line to these points using the Hough Transform. For simplicity, use the linear version of the Hough Transform: $y = ax + b$.